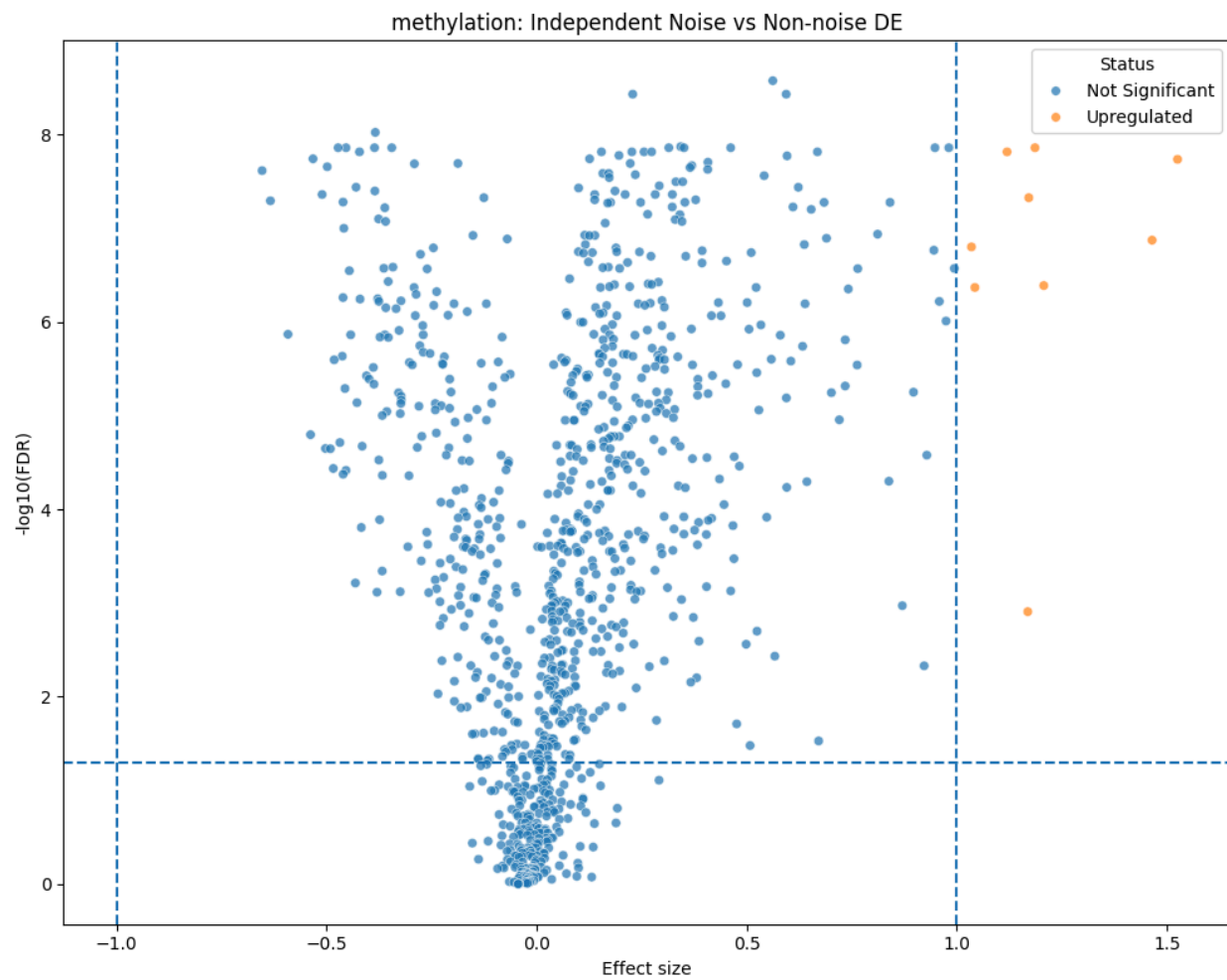
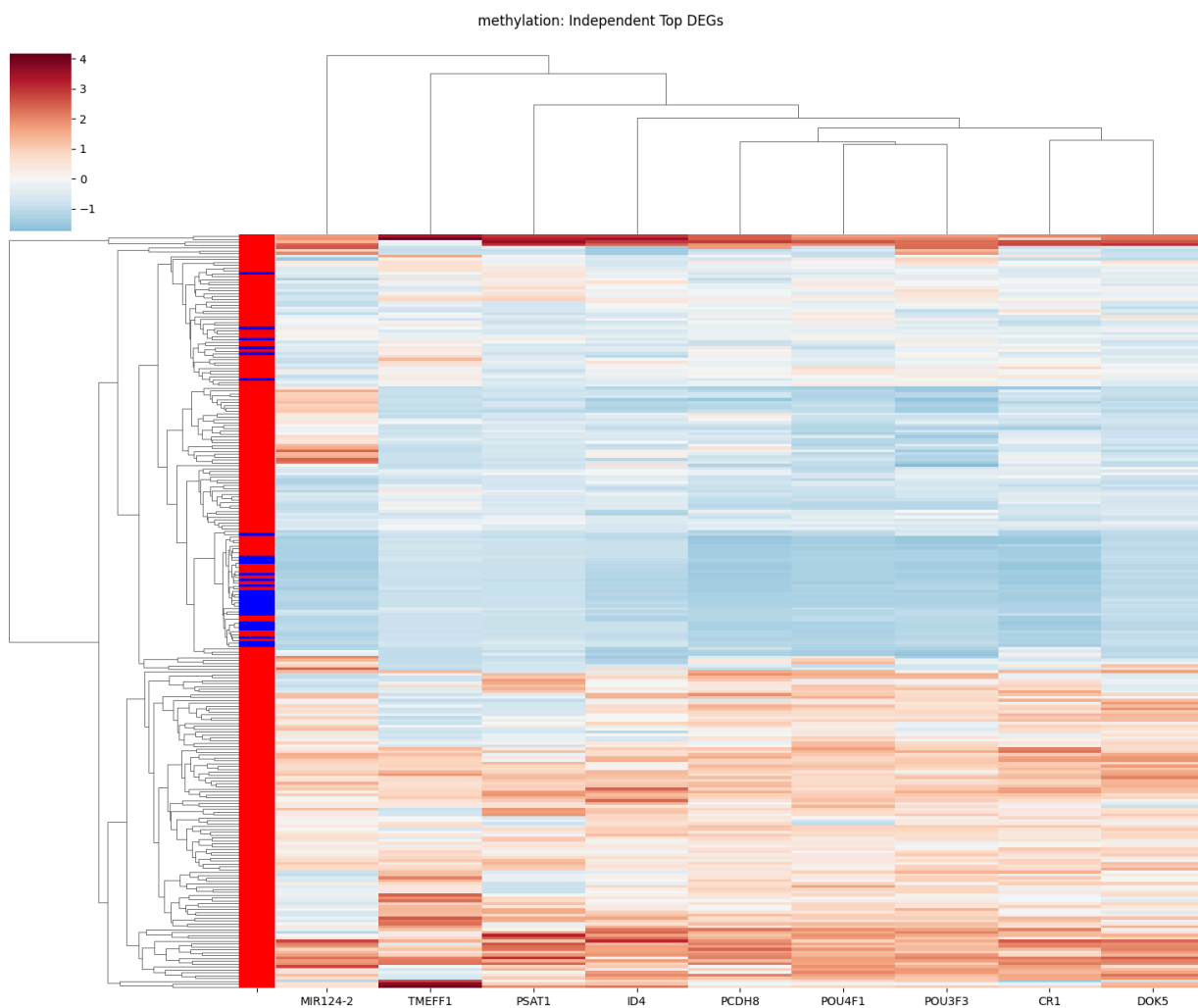






```
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=====
KNOWN CANCER-ASSOCIATED GENE ANNOTATION
=====
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```

Total annotated records: 4
Unique genes identified: 2

Category-wise counts:
Known_Biological_Category
Cancer Epigenetic Markers 1
Tumor Suppressor 1
PI3K_AKT 1
RAS_MAPK 1
Name: count, dtype: int64

Annotated genes:
Gene Known_Cancer_Gene Known_Biological_Category
MIR124-2 MIR124-2 Cancer Epigenetic Markers
DOK5 DOK5 Tumor Suppressor
DOK5 DOK5 PI3K_AKT
DOK5 DOK5 RAS_MAPK

```
=====
=====
Annotation saved to:
/content/methylation_Reviewer_Corrected_Results/Known_Cancer_Gene_Annotation.csv
=====
=====
```

Genes for GO: 9

	Gene_set \
0	GO_Biological_Process_2026
1	GO_Biological_Process_2026
2	GO_Biological_Process_2026
3	GO_Biological_Process_2026
4	GO_Biological_Process_2026
5	GO_Biological_Process_2026
6	GO_Biological_Process_2026
7	GO_Biological_Process_2026
8	GO_Biological_Process_2026
9	GO_Biological_Process_2026
10	GO_Biological_Process_2026
11	GO_Biological_Process_2026
12	GO_Biological_Process_2026
13	GO_Biological_Process_2026
14	GO_Biological_Process_2026
15	GO_Biological_Process_2026
16	GO_Biological_Process_2026
17	GO_Biological_Process_2026
18	GO_Biological_Process_2026
19	GO_Biological_Process_2026

	Term Overlap	P-value \
0	Negative Regulation of Programmed Cell Death (... 3/410	0.000655
1	Metanephric Distal Tubule Development (GO:0072...	1/5 0.002248
2	Negative Regulation of Immunoglobulin Producti...	1/5 0.002248
3	Regulation of Neurotrophin TRK Receptor Signal...	1/5 0.002248
4	Peripheral Nervous System Neuron Differentiati...	1/5 0.002248
5	Regulation of Plasma Cell Differentiation (GO:...	1/5 0.002248
6	Negative Regulation of Cell Killing (GO:0031342)	1/5 0.002248
7	Negative Regulation of Complement-Dependent Cy...	1/5 0.002248
8	Negative Regulation of B Cell Differentiation ...	1/5 0.002248
9	ATP Export (GO:1904669)	1/6 0.002697
10	Peripheral Nervous System Neuron Development (... 1/6	0.002697
11	Trigeminal Nerve Development (GO:0021559)	1/6 0.002697
12	Positive Regulation of Endopeptidase Activity ...	1/6 0.002697
13	Negative Regulation of Humoral Immune Response...	1/7 0.003146
14	Metanephric Nephron Tubule Development (GO:007...	1/7 0.003146
15	Regulation of Complement Activation, Alternati...	1/7 0.003146
16	Negative Regulation of Complement Activation, ...	1/7 0.003146
17	Regulation of Complement Activation, Classical...	1/8 0.003595
18	Positive Regulation of Complement Activation (... 1/8	0.003595
19	Negative Regulation of Endopeptidase Activity ...	1/8 0.003595

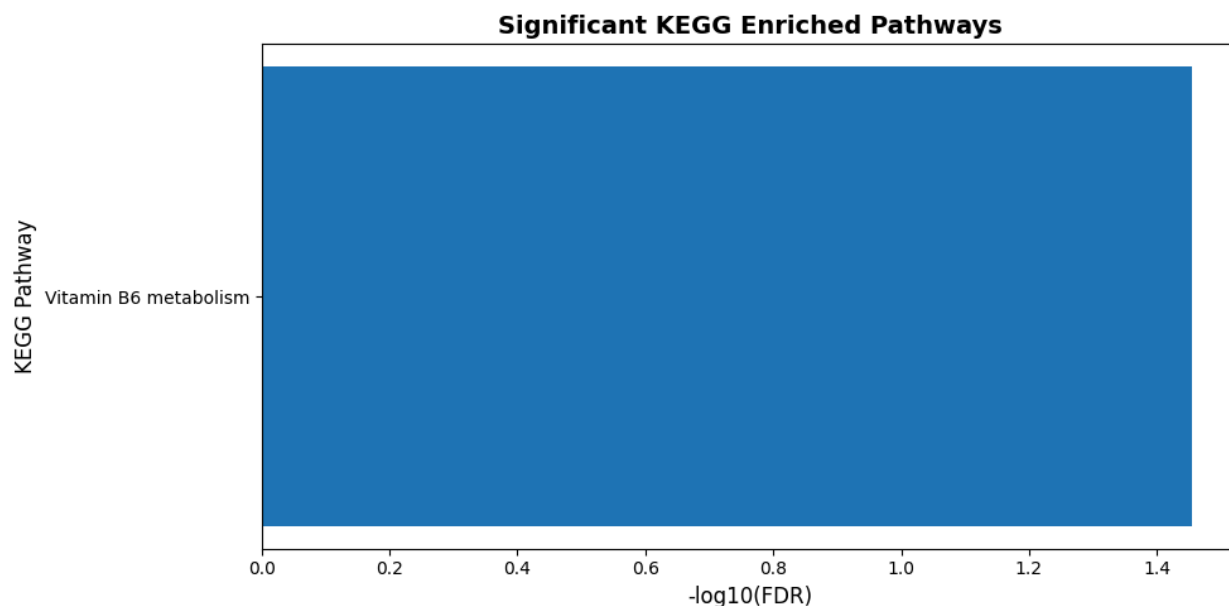
	Adjusted P-value	Old P-value	Old Adjusted P-value	Odds Ratio \
0	0.020049	0	0	24.058968
1	0.020049	0	0	624.593750
2	0.020049	0	0	624.593750
3	0.020049	0	0	624.593750
4	0.020049	0	0	624.593750
5	0.020049	0	0	624.593750
6	0.020049	0	0	624.593750
7	0.020049	0	0	624.593750
8	0.020049	0	0	624.593750
9	0.020049	0	0	499.650000
10	0.020049	0	0	499.650000
11	0.020049	0	0	499.650000
12	0.020049	0	0	499.650000
13	0.020049	0	0	416.354167
14	0.020049	0	0	416.354167
15	0.020049	0	0	416.354167
16	0.020049	0	0	416.354167
17	0.020049	0	0	356.857143
18	0.020049	0	0	356.857143
19	0.020049	0	0	356.857143

	Combined Score	Genes
0	176.362965	PSAT1;POU4F1;POU3F3
1	3808.549663	POU3F3
2	3808.549663	CR1
3	3808.549663	DOK5
4	3808.549663	POU4F1
5	3808.549663	CR1
6	3808.549663	CR1
7	3808.549663	CR1
8	3808.549663	CR1
9	2955.689739	CR1
10	2955.689739	POU4F1
11	2955.689739	POU4F1
12	2955.689739	CR1
13	2398.853181	CR1
14	2398.853181	POU3F3
15	2398.853181	CR1
16	2398.853181	CR1
17	2008.476466	CR1
18	2008.476466	CR1
19	2008.476466	CR1

Gene_set	Term \
0 KEGG_2021_Human	Vitamin B6 metabolism
1 KEGG_2021_Human	Glycine, serine and threonine metabolism
2 KEGG_2021_Human	Cysteine and methionine metabolism
3 KEGG_2021_Human	Malaria
4 KEGG_2021_Human	Legionellosis
5 KEGG_2021_Human	Leishmaniasis
6 KEGG_2021_Human	Complement and coagulation cascades
7 KEGG_2021_Human	TGF-beta signaling pathway
8 KEGG_2021_Human	Hematopoietic cell lineage
9 KEGG_2021_Human	Signaling pathways regulating pluripotency of ...
10 KEGG_2021_Human	Tuberculosis
11 KEGG_2021_Human	Neutrophil extracellular trap formation
12 KEGG_2021_Human	MicroRNAs in cancer

	Overlap	P-value	Adjusted P-value	Old P-value	Old Adjusted P-value \
0	1/6	0.002697	0.035064	0	0
1	1/40	0.017860	0.063103	0	0
2	1/50	0.022281	0.063103	0	0
3	1/50	0.022281	0.063103	0	0
4	1/57	0.025364	0.063103	0	0
5	1/77	0.034128	0.063103	0	0
6	1/85	0.037613	0.063103	0	0
7	1/94	0.041521	0.063103	0	0
8	1/99	0.043686	0.063103	0	0
9	1/143	0.062552	0.081317	0	0
10	1/180	0.078159	0.088747	0	0
11	1/189	0.081920	0.088747	0	0
12	1/310	0.131181	0.131181	0	0

	Odds Ratio	Combined Score	Genes
0	499.650000	2955.689739	PSAT1
1	63.948718	257.405594	PSAT1
2	50.872449	193.520870	PSAT1
3	50.872449	193.520870	CR1
4	44.497768	163.503036	CR1
5	32.754934	110.634590	CR1
6	29.623512	97.176861	CR1
7	26.744624	85.089291	ID4
8	25.373724	79.438007	CR1
9	17.472711	48.430150	ID4
10	13.835196	35.266017	CR1
11	13.166888	32.943637	CR1
12	7.961974	16.172185	MIR124-2



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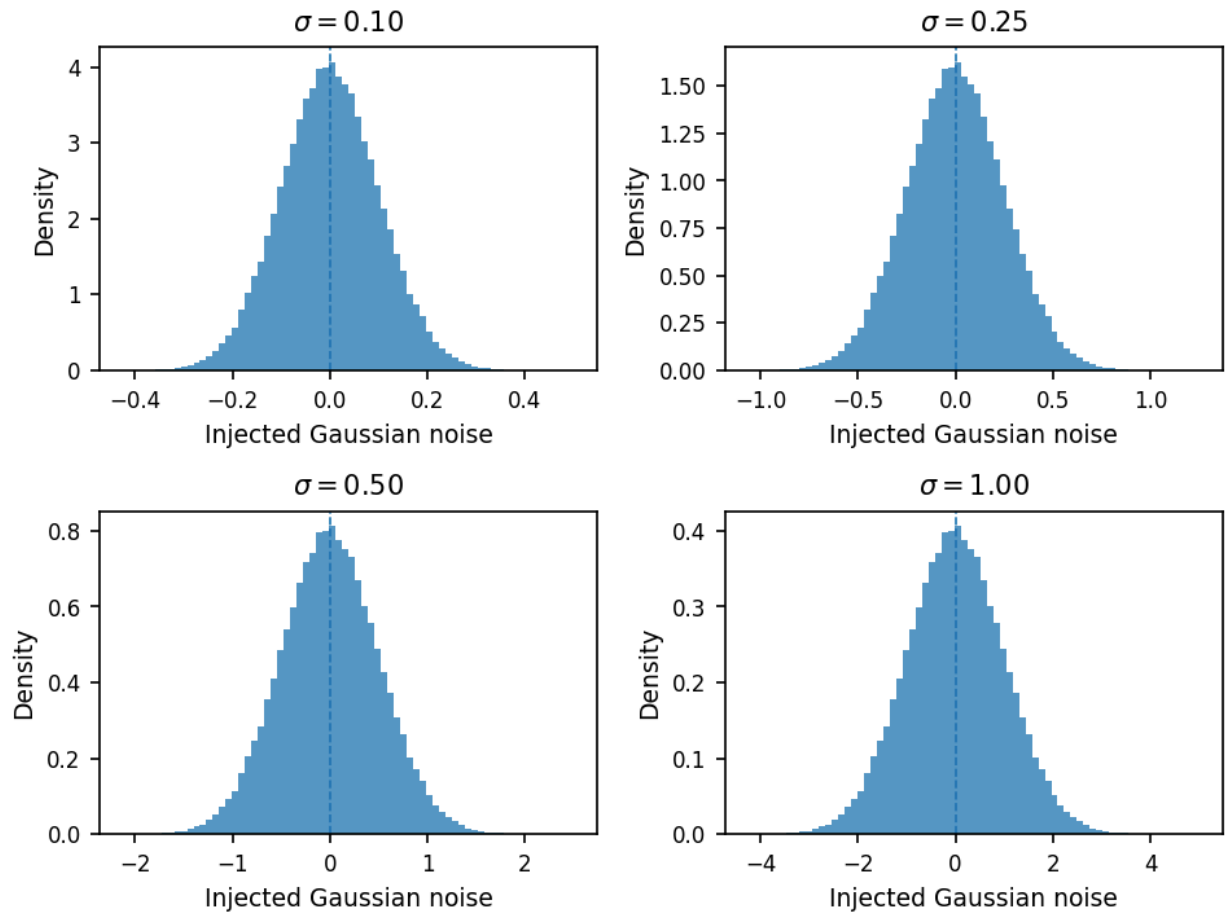
DETAILED RESULTS

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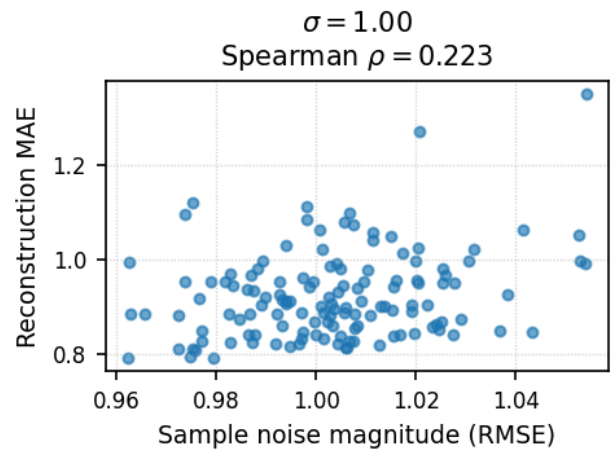
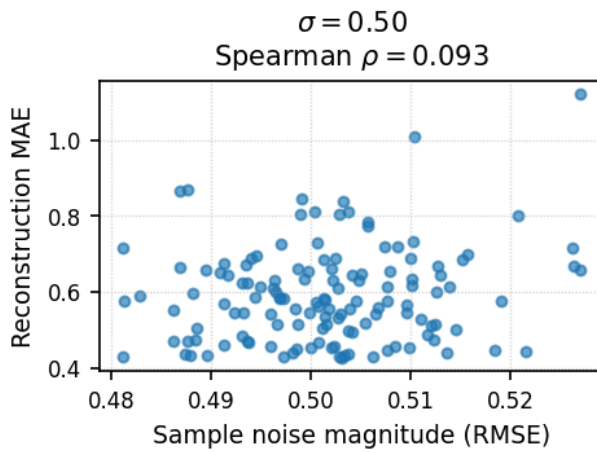
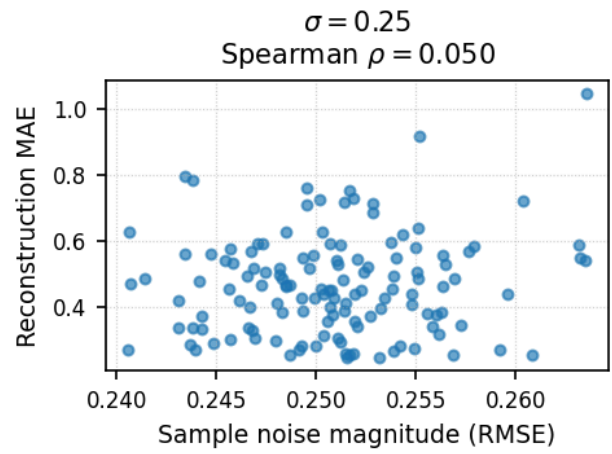
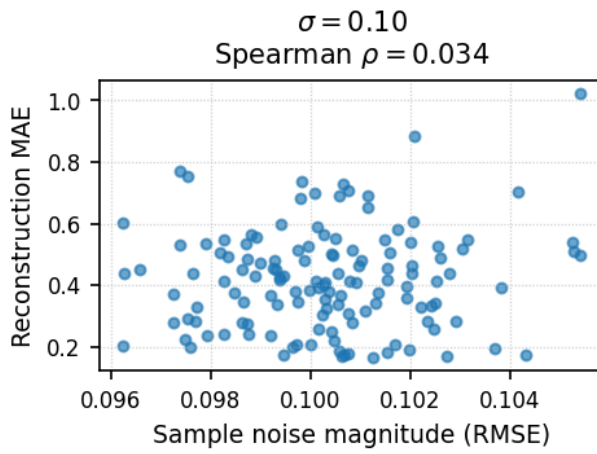
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Noise_Level	Actual_Noise_STD	Mean_MAE	Median_MAE	SD_MAE	Mean_RMSE
Median_RMSE	SD_RMSE	Max_MAE	Max_RMSE		
0.10	0.100346	0.420496	0.415140	0.163448	0.555918
0.25	0.250865	0.467771	0.456432	0.149519	0.609215
0.50	0.501729	0.594221	0.578967	0.125001	0.759083
1.00	1.003458	0.923672	0.904723	0.092507	1.164253

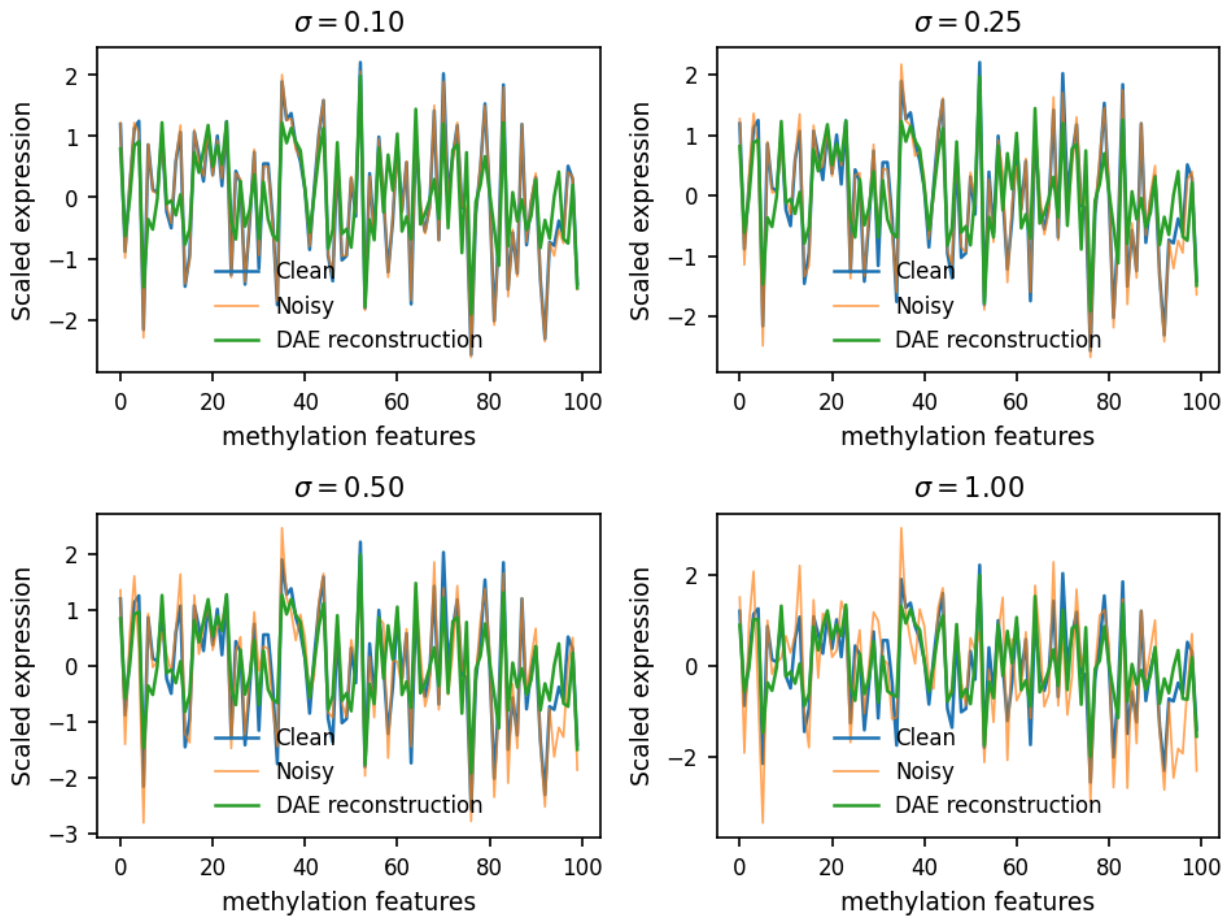
Distribution of Synthetic Technical Noise



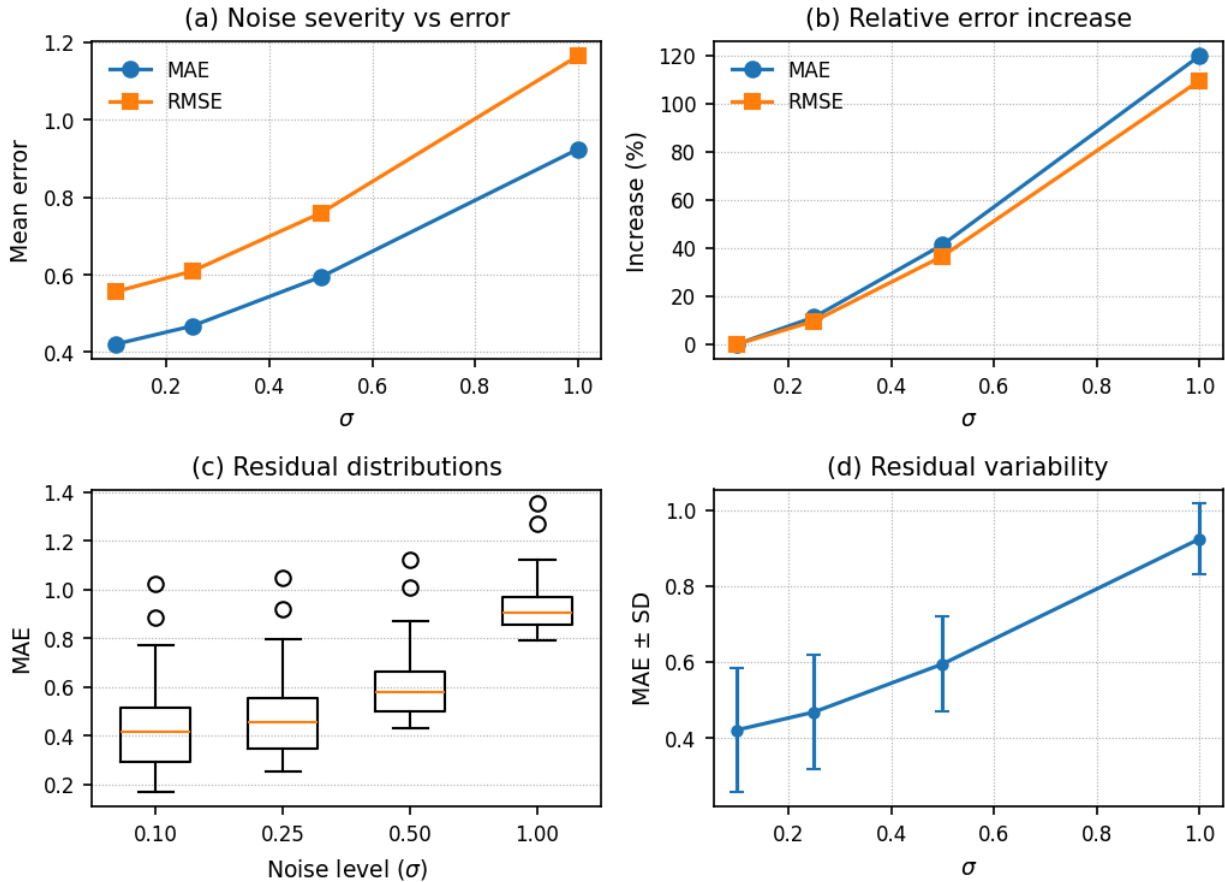
Injected Noise Magnitude vs Reconstruction Residual



Representative methylation Signal Before and After Denoising



Synthetic Gaussian Technical-Noise Analysis of methylation



M4 — FINAL SYNTHETIC TECHNICAL-NOISE ANALYSIS

Noise $\sigma = 0.10$

Actual noise STD : 0.100346

Mean MAE : 0.420496

Median MAE : 0.415140

Mean RMSE : 0.555918

Median RMSE : 0.549033

Noise $\sigma = 0.25$

Actual noise STD : 0.250865

Mean MAE : 0.467771

Median MAE : 0.456432

Mean RMSE : 0.609215

Median RMSE : 0.595510

Noise $\sigma = 0.50$

Actual noise STD : 0.501729

Mean MAE : 0.594221

Median MAE : 0.578967

Mean RMSE : 0.759083

Median RMSE : 0.737699

Noise $\sigma = 1.00$

Actual noise STD : 1.003458

Mean MAE : 0.923672

Median MAE : 0.904723

Mean RMSE : 1.164253

Median RMSE : 1.139065